

INDEX

S.N.	Name of Laboratory Work	page
1.	Static and Auto configuration of IPV6 address using Packet Tracer	2
2.	IPV6 Header and Packet Analysis (Neighbor/Router Solicitation/Discovery)	7
3.	Static Routing in IPV6	13
4.	Dynamic Routing in IPV6: OSPFv3 Implementation	18

Lab 1: Static and Auto configuration of IPV6 address using Packet Tracer

Objectives

- To understand the concept of IPV6 and its address configuration
- Configure EUI-64 Format Global Unicast Address
- Configure Manual Global Unicast Address
- Auto IPv6 Address Configuration

Theory

An IPv6 address is 128-bit long and uses 8 groups of 4 hexadecimal digits separated by colons.

An IPv6 address looks like this:

FE80:0A3B: 0002:4B3C:0F6D: 3D40:FFC5:005A

Long version

2002:0000:3D12:25AF: 2788:00AB: 63AF:002C

Shortened version

2002:0:3D12:25AF:2788: AB: 63AF:2C

Long version FF02:0000:3D12:0000:0000: 0000:63AF:002C

Shortened version FF02:0:3D12::63AF:2C

IPv6 address types

In IPv4, we have 3 types of addresses: unicast, multicast and broadcast. In IPv6, we no longer have broadcast. Broadcast addresses have been eliminated and replaced with anycast and multicast addresses. So, for IPv6, we have: unicast, multicast and anycast.

1. Unicast address: Identifies a single interface. Packets addressed to a unicast address are delivered to a single interface
2. Multicast address: Identifies a specific group of hosts. Just as for IPv4, packets sent to a multicast address are delivered to all interfaces identified by the multicast address.
Multicast IPv6 addresses always start with FF

3. Anycast address: Like a multicast address, an anycast address identifies multiple interfaces, but there's a difference: packets sent to an anycast address are delivered to one address- the address which is nearest to the source in terms of routing distance.

Procedure

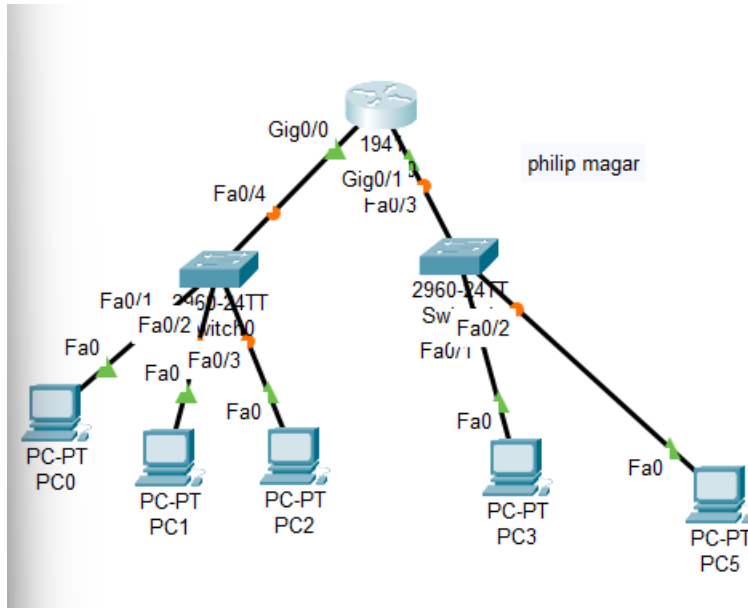


Figure: Network Topology

In Router:

Router>

Router>show ip interface

Router>enable

Router # configure terminal Router (config)#ipv6 unicast-routing

Router (config)#interface gig 0/0

Router (config-if) #ipv6 enable

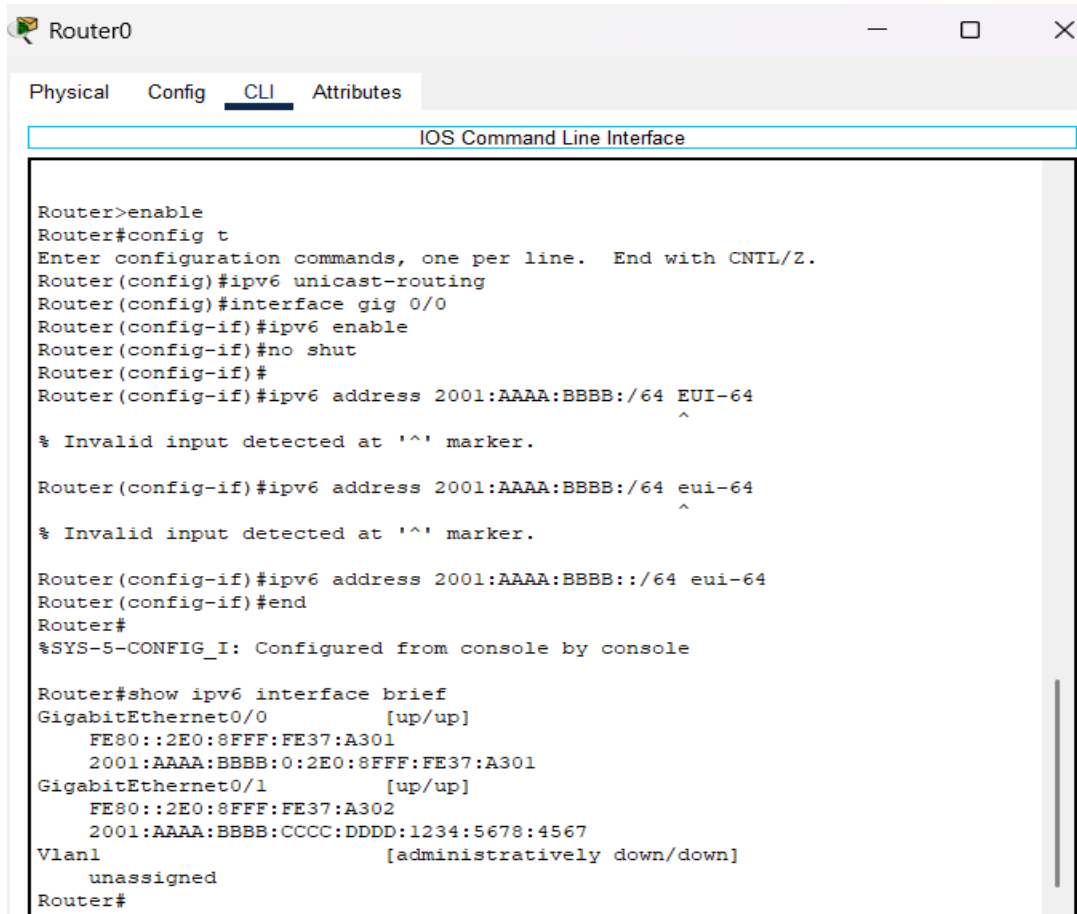
We assign unicast address as following

Router (config-if) #ipv6 address 2001: AAAA: BBBB:/64 EUI-64

Router (config-if) #no shutdown

Router (config-if) #end

Router #show ipv6 interface brief



The screenshot shows a window titled 'Router0' with tabs for 'Physical', 'Config', 'CLI', and 'Attributes'. The 'CLI' tab is active, displaying the 'IOS Command Line Interface'. The terminal output shows the following sequence of commands and responses:

```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ipv6 unicast-routing
Router(config)#interface gig 0/0
Router(config-if)#ipv6 enable
Router(config-if)#no shut
Router(config-if)#
Router(config-if)#ipv6 address 2001:AAAA:BBBB:/64 EUI-64
                                     ^
% Invalid input detected at '^' marker.

Router(config-if)#ipv6 address 2001:AAAA:BBBB:/64 eui-64
                                     ^
% Invalid input detected at '^' marker.

Router(config-if)#ipv6 address 2001:AAAA:BBBB::/64 eui-64
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ipv6 interface brief
GigabitEthernet0/0      [up/up]
    FE80::2E0:8FFF:FE37:A301
    2001:AAAA:BBBB:0:2E0:8FFF:FE37:A301
GigabitEthernet0/1      [up/up]
    FE80::2E0:8FFF:FE37:A302
    2001:AAAA:BBBB:CCCC:DDDD:1234:5678:4567
Vlan1                   [administratively down/down]
    unassigned
Router#
```

Figure: Basic router configuration with eui-64

Router# configure terminal

Router (config)# interface gig0/1

Router (config-if) #ipv6 enable

**Router (config-if) #ipv6 address2001: AAAA: BBBB: CCCC: DDDD:
1234:5678:4567/64**

Router (config-if) #end

Router # show ipv6 interface brief

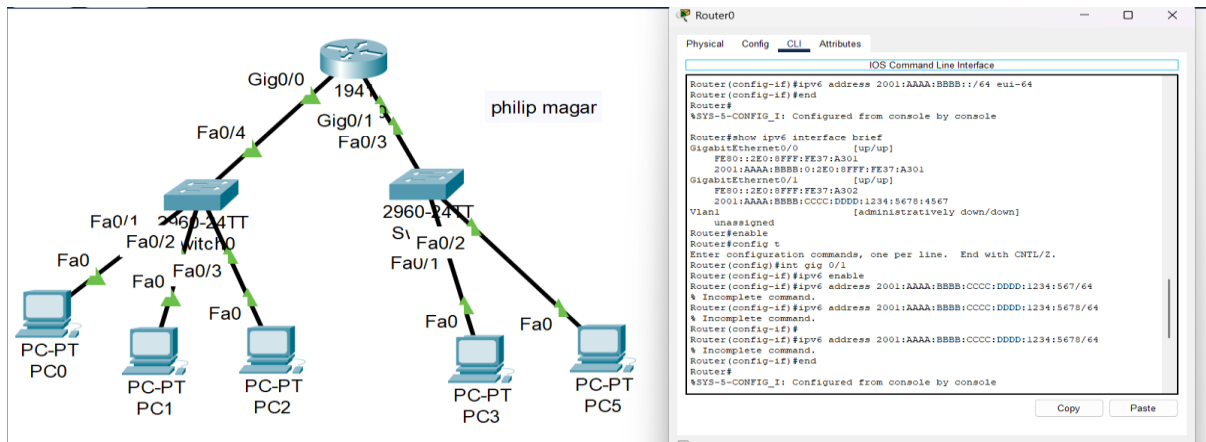


Figure: Interface gig0/1 configured with static IP

In PC, we can assign IP either statically or using auto-configuration.

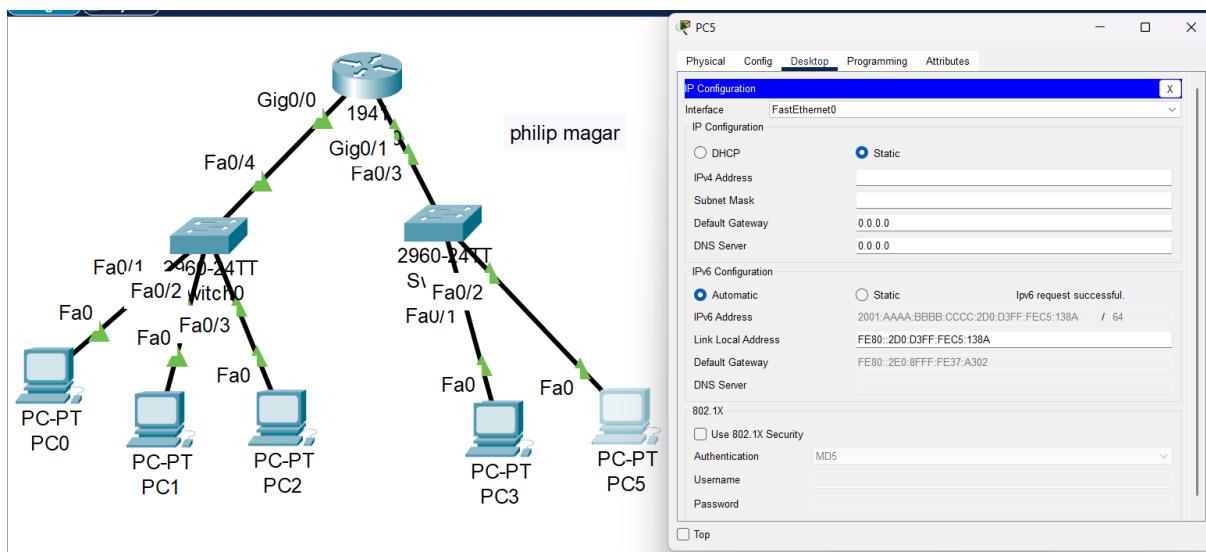
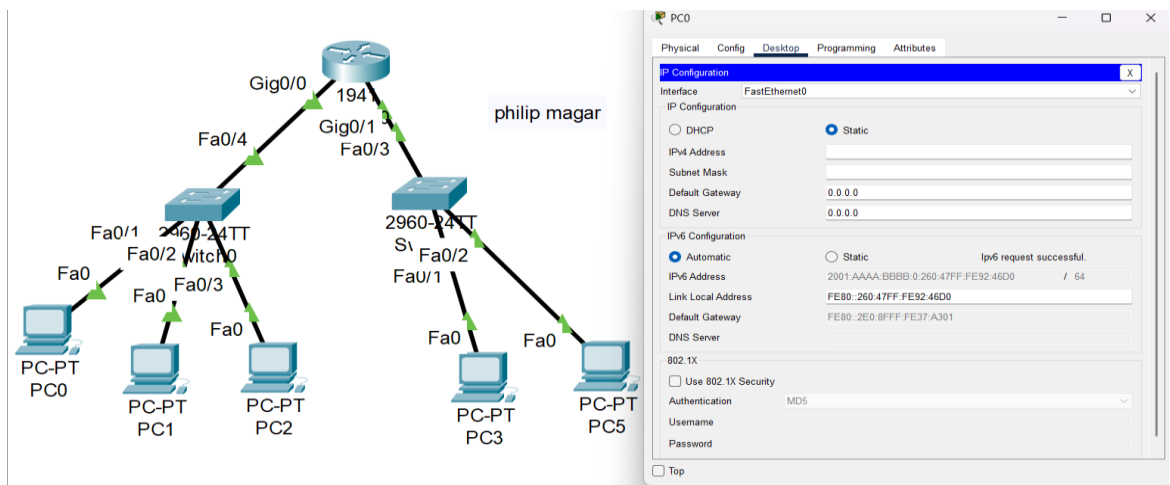
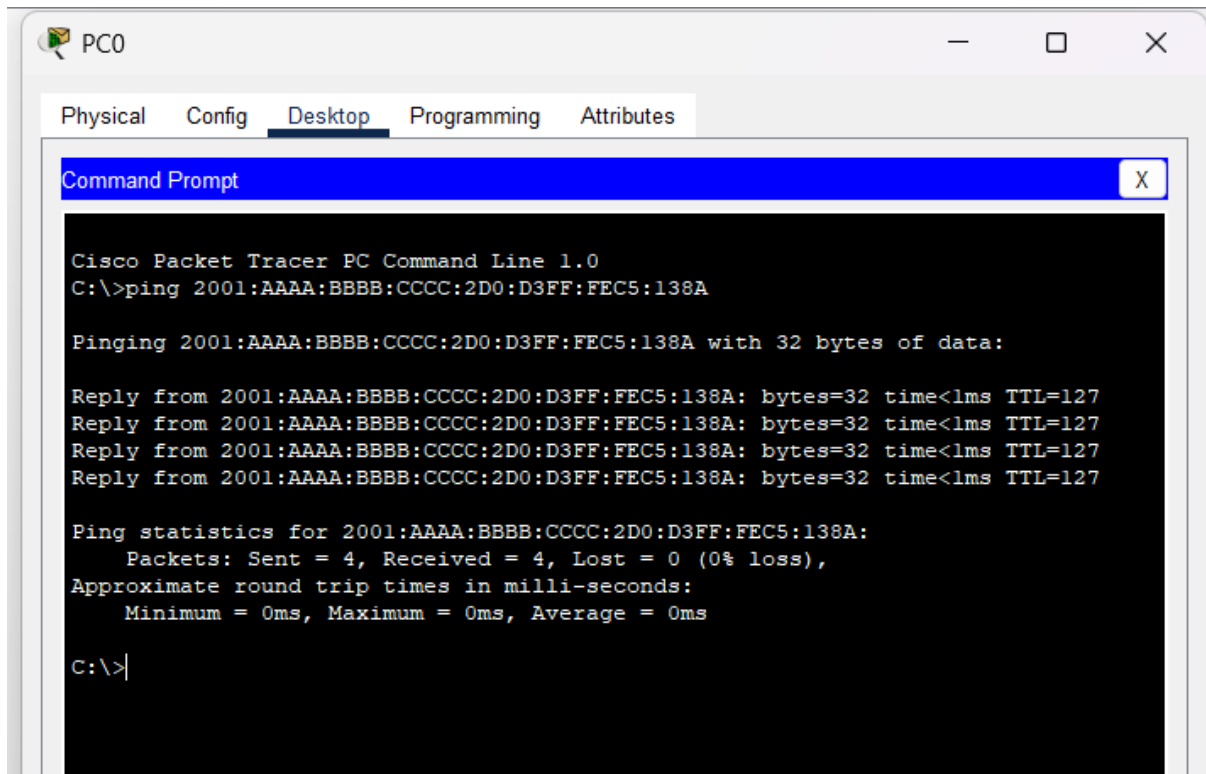


Figure: PC0 and PC5 configured with automatic-configuration



Auto Configuration of IPv6 Address

IPv6 Addresses can be configured automatically. For IPv6 Auto configuration, we will use “ipv6 address autoconfig” command. We have auto configured in PCs in above topology but we can do the same thing in router interface as well.

Example:

Router(config)# interface gig0/2

Router(config_if) #ipv6 address autoconfig

Router(config_if) #end

Lab 2: IPV6 Header and Packet Analysis (Neighbor/Router Solicitation/Discovery)

Objectives

- To understand the packet format of IPV6
- To know how neighbor discovery protocol works using ICMPV6 protocol

Theory

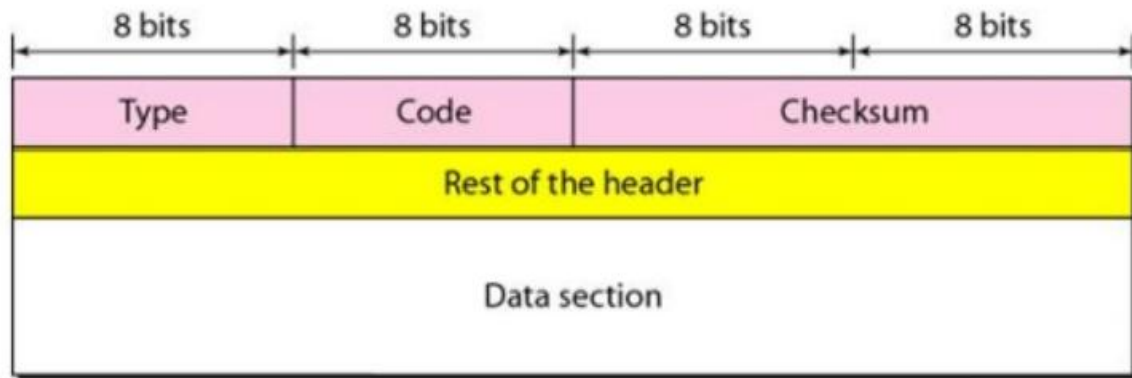


Figure: ICMPV6 general Message Format

Neighbor Discovery

Both host and routers use ND to determine whether a link-local or global unicast address it is about to use is already used by another device (DAD: Duplicate Address Detection).

Neighbor Discovery Messages

- Router Solicitation (ICMPv6 type 133)
- Router Advertisement (ICMPv6 type 134)
- Neighbor Solicitation (ICMPv6 type 135)
- Neighbor Advertisement (ICMPv6 type 136)

Procedure

Router>

Router>enable

Router # configure terminal

Router (config) # hostname philip

philip(config)#interface gig 0/0

philip(config-if) #ipv6 address 2001: ABCD:1234::1/64

The following command is link local used only in local network. It is compulsory to have link local for every router on the network in IPV6

philip(config-if) #ipv6 address FE80::1 link-local

philip(config-if) #no shutdown

philip(config-if) #exit

philip(config)#ipv6 unicast-routing

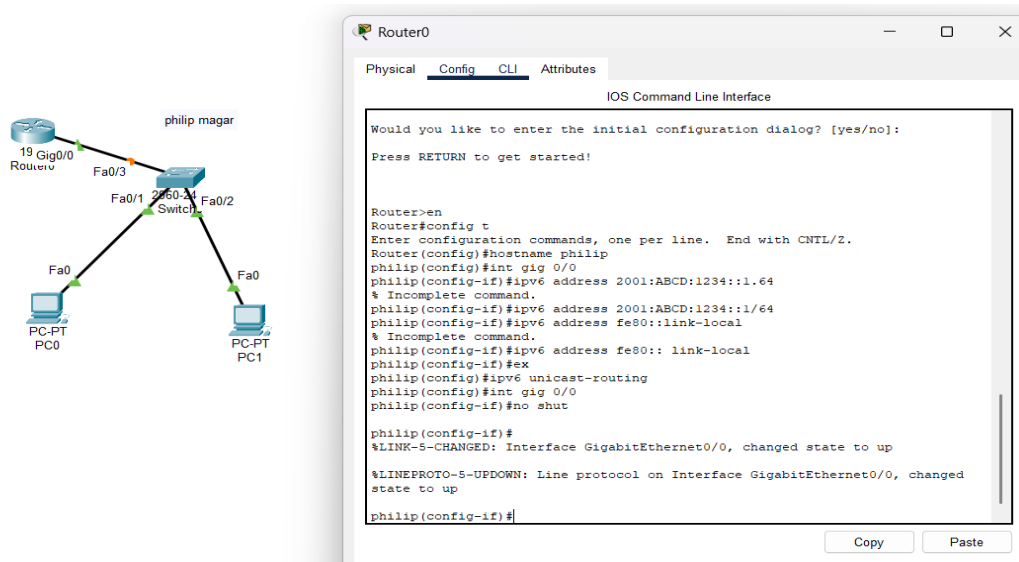


Figure: IPV6 address configuration

Now go to PC and select automatic in IPV6. As soon as we select automatic, we can see a packet in the PC, we can see neighbor discovery protocol packet being generated, that is the router solicitation message.

When we press play button, it moves to switch and if we press the packet, we can see Ethernet and IPV6 information. Now router takes the message and does router advertisement and router solicitation and PC gets IP address and prefix.

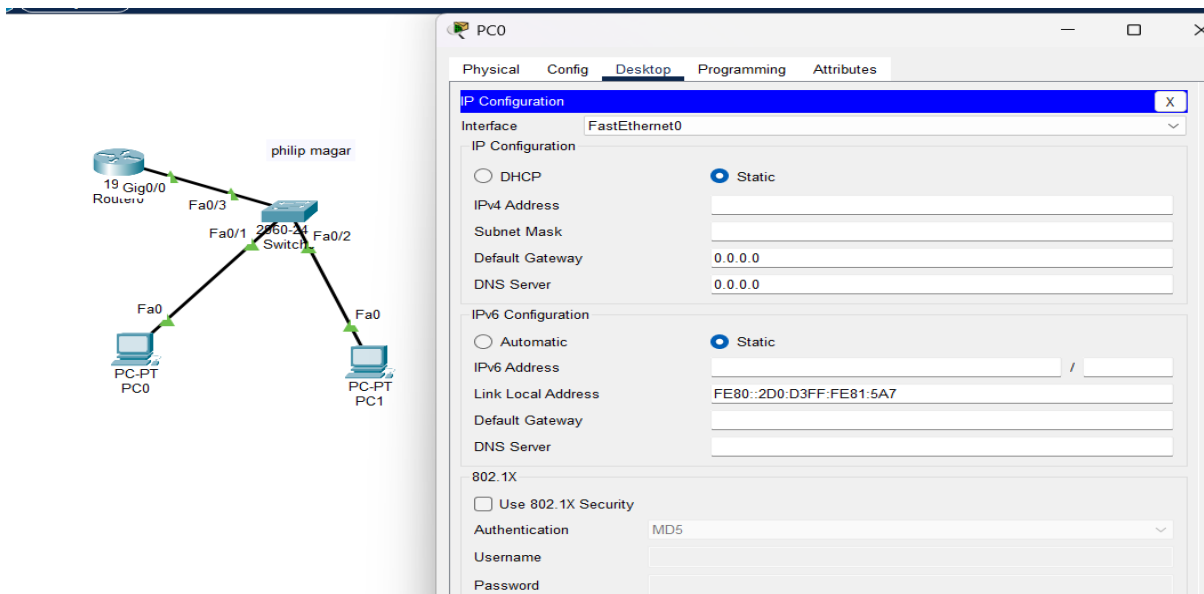


Figure: PC generating Router Solicitation message

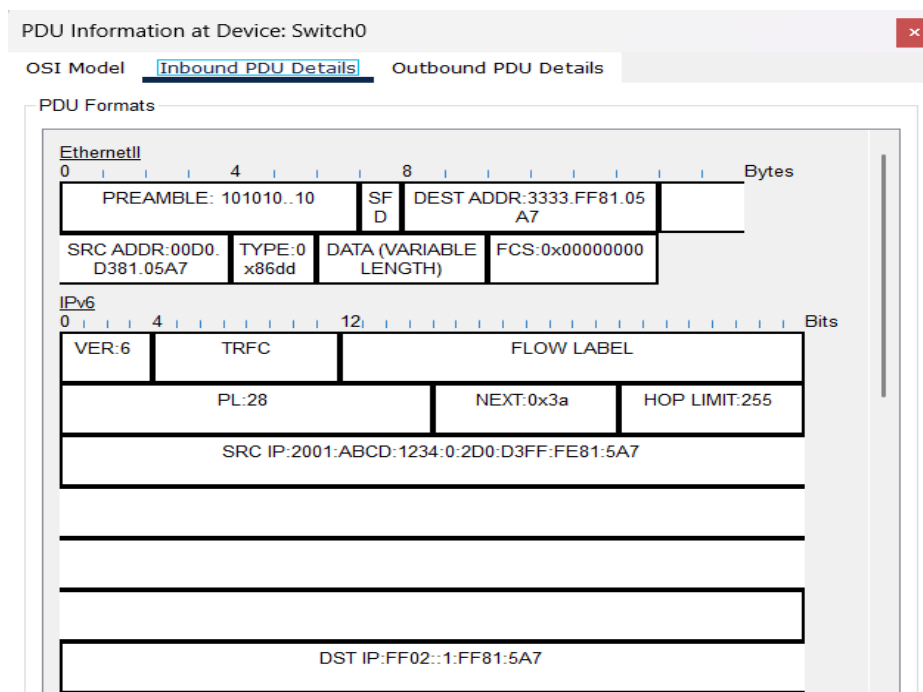


Figure: Ethernet and IPV6

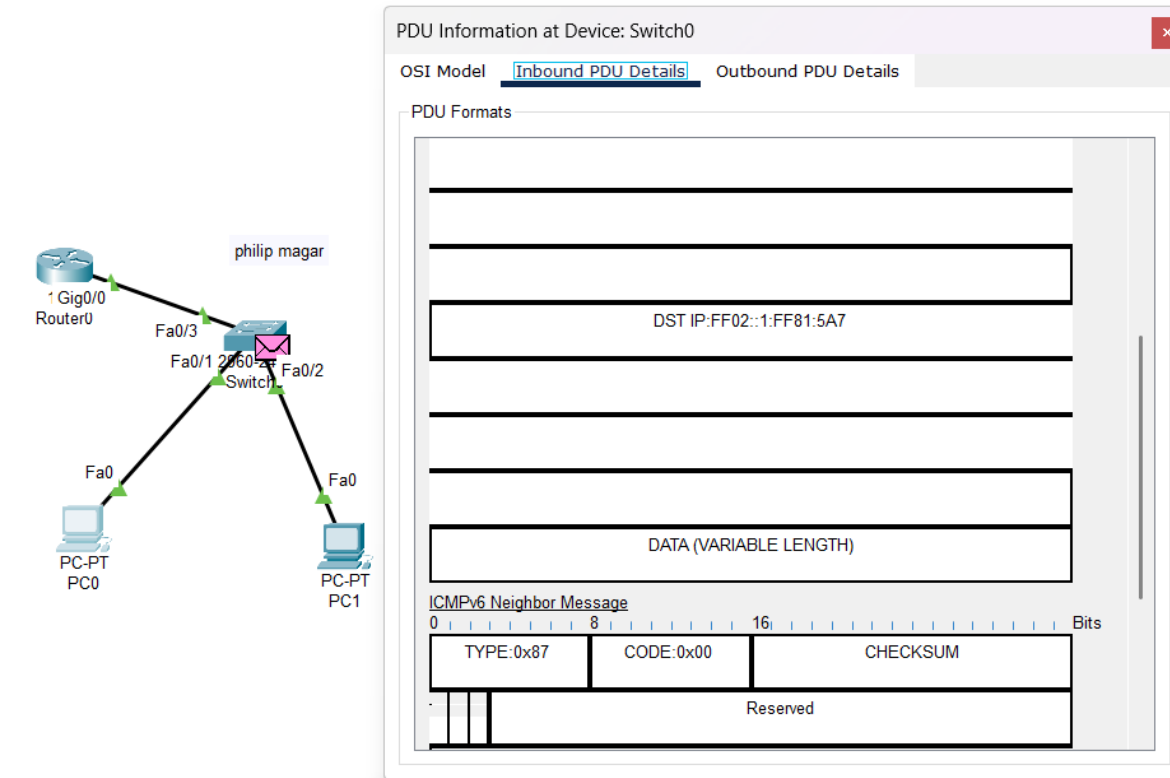


Figure: Router solicitation message

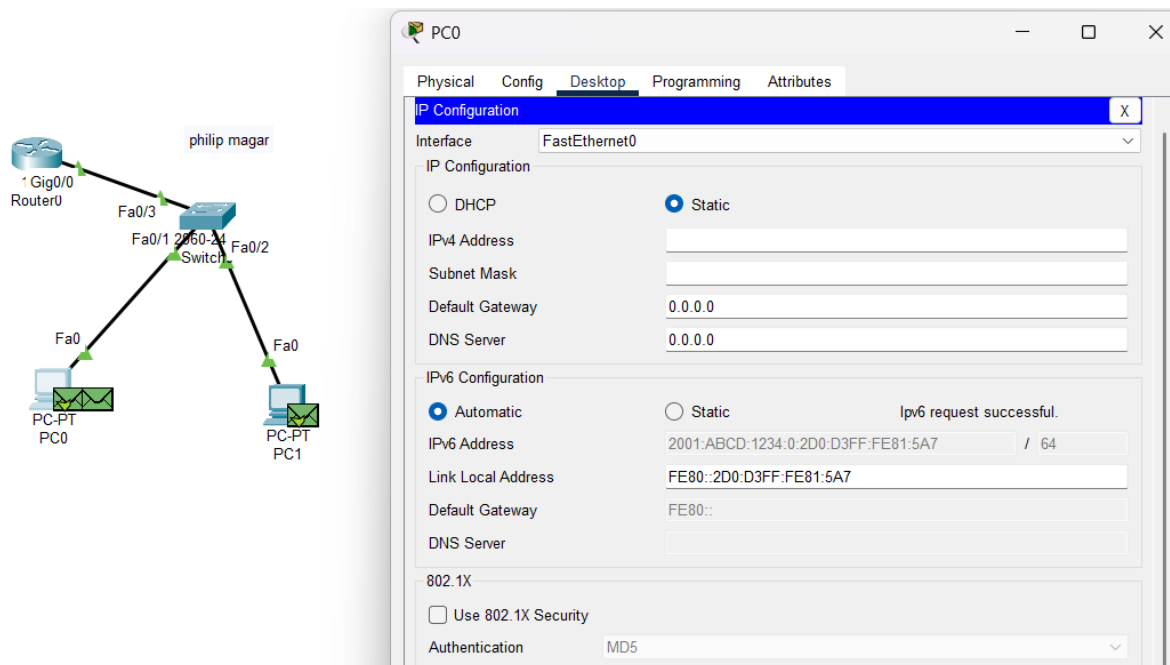


Figure: IPV6 auto configuration in PC

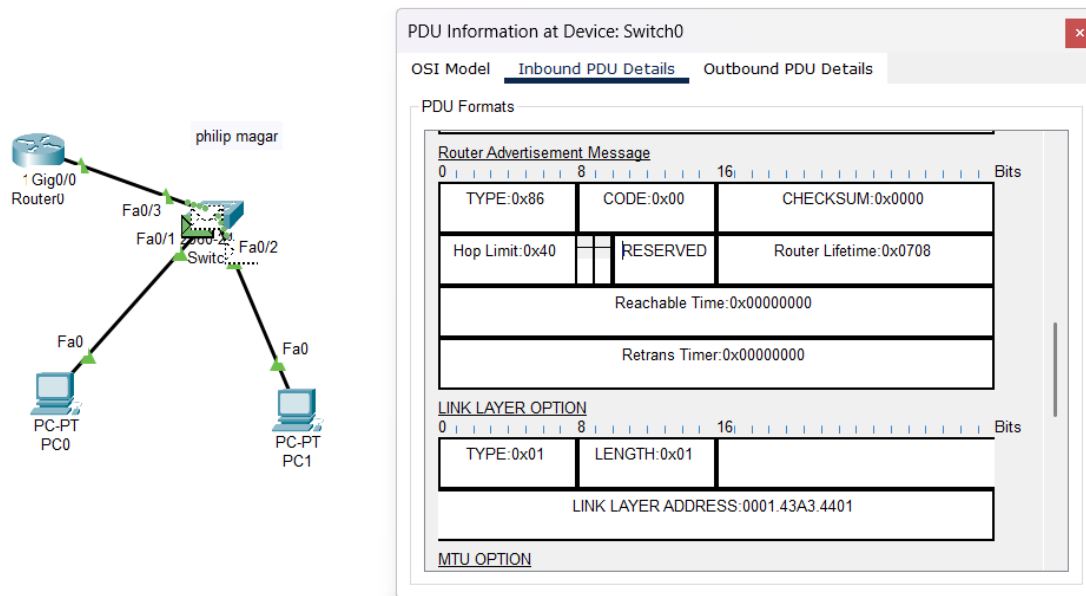


Figure: Router Advertisement message

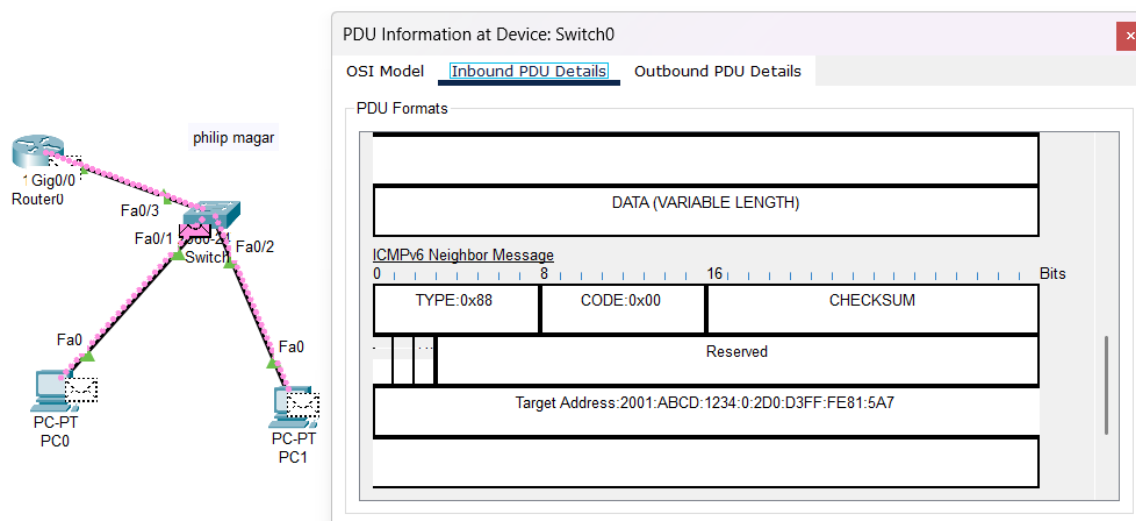
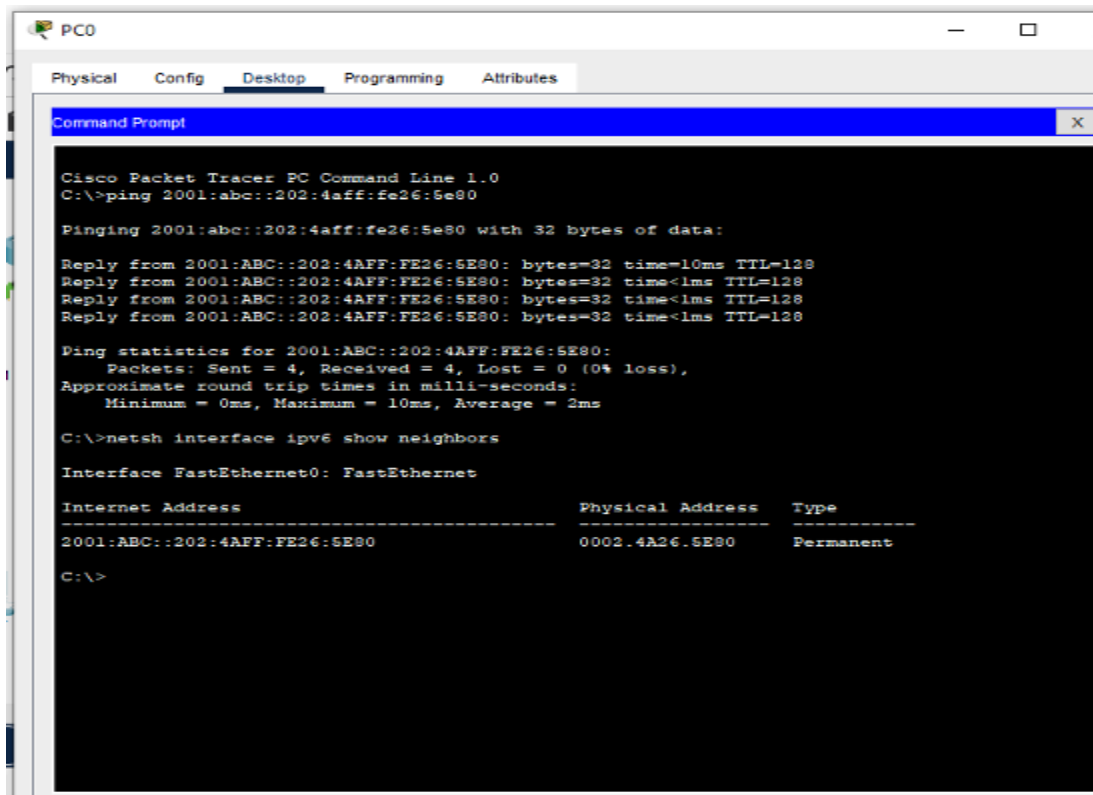


Figure: Neighbor Discovery Packet

Now if we configure both PC with IP and sent packet from one PC to other, we can see neighbor discovery packet as above We didn't use ARP in IPV6, so to see MAC address of the connected PC, go to command prompt of PC and type

C:>netsh interface ipv6 show neighbors



Lab 3: Static Routing

Objective To understand how system admin can configure static routing in the router

Theory

Static routes are manually entered into a router or host. An administrator must know the internetwork layout and the paths that exist between networks.

Procedure

In the diagram below, we have used two routers to connect two different networks.

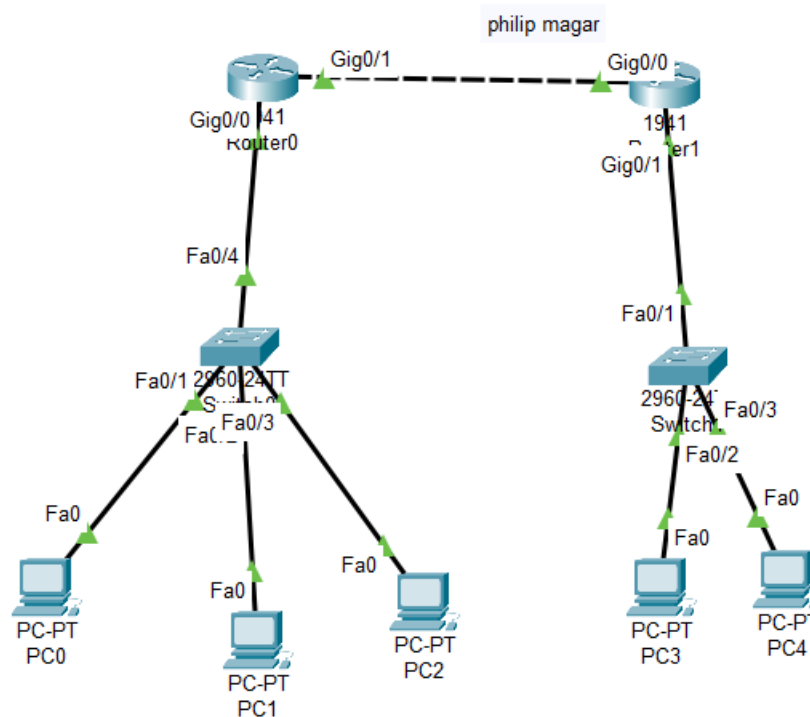


Figure: Network Topology

Router>

Router>enable

Router # configure terminal

Router (config)#interface gig 0/0

Router (config-if) #ipv6 address 2001::1/64

Router (config)#ipv6 unicast-routing

Router (config-if) #no shutdown

Router (config-if) #interface gig0/1

Router (config-if) #ipv6 address 2000::1/64

Router (config-if) #no shutdown

Similarly, we repeat for next route with IPV6 address as 2002::1/64 and 2000::2/64

For static routing.

Router (config)#ipv6 route 2002: :/64 2000::2

Similarly for next router

Router (config)#ipv6 route 2001: :/64 2000::1

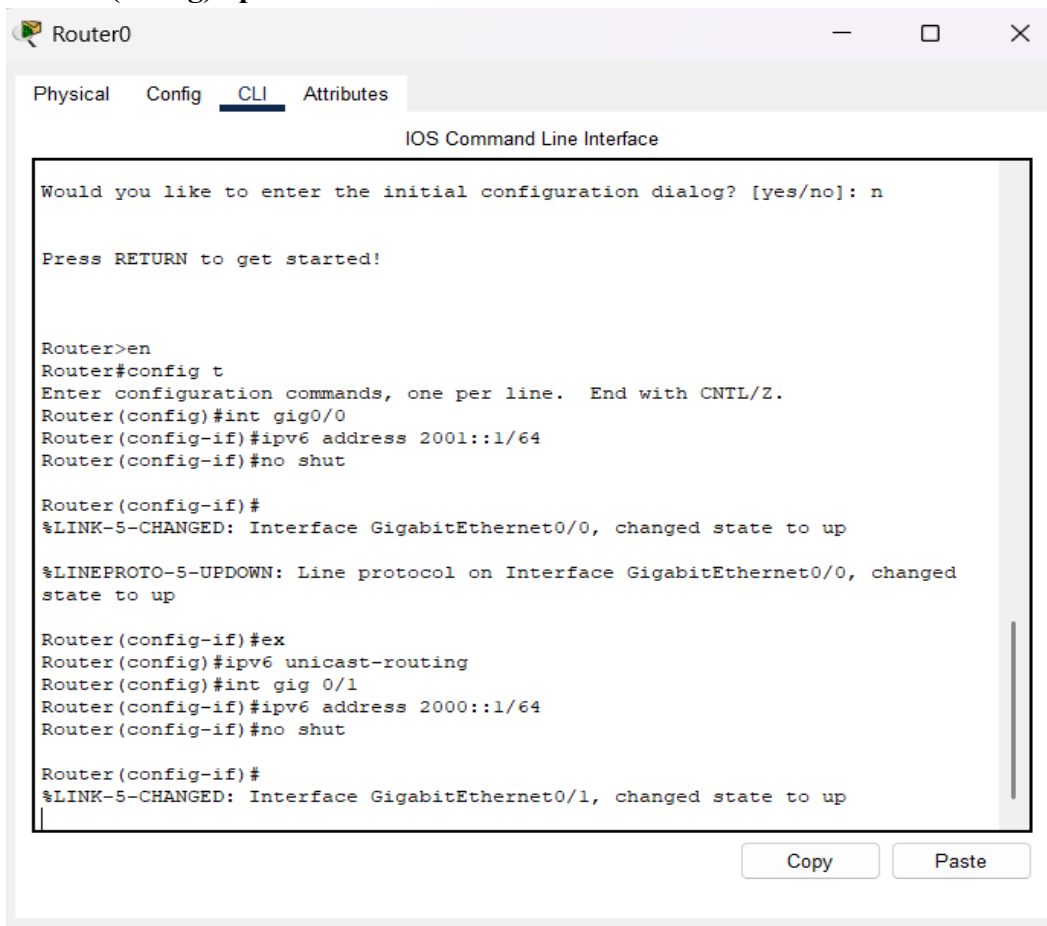


Figure: IPV6 address configuration

Router# show ipv6 route

Router# show ipv6 protocol

Router# show ipv6 neighbors

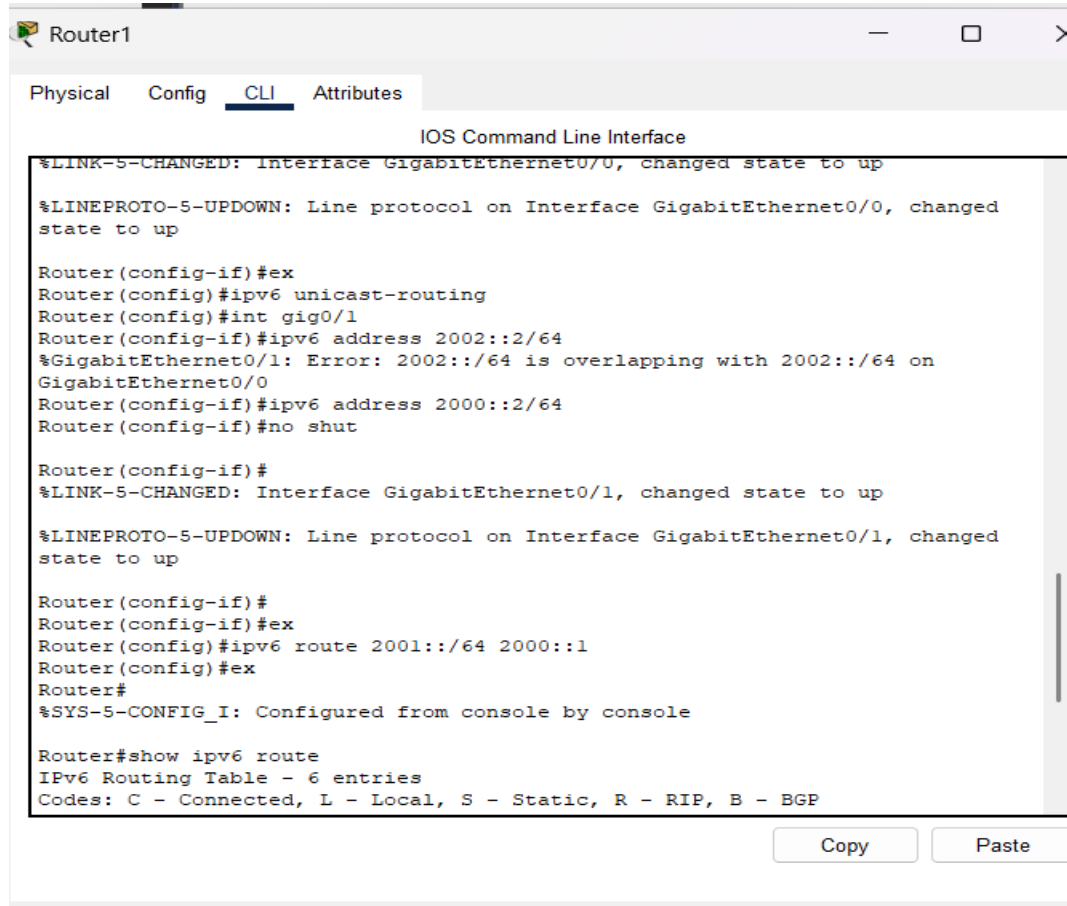


Figure: Defining static route

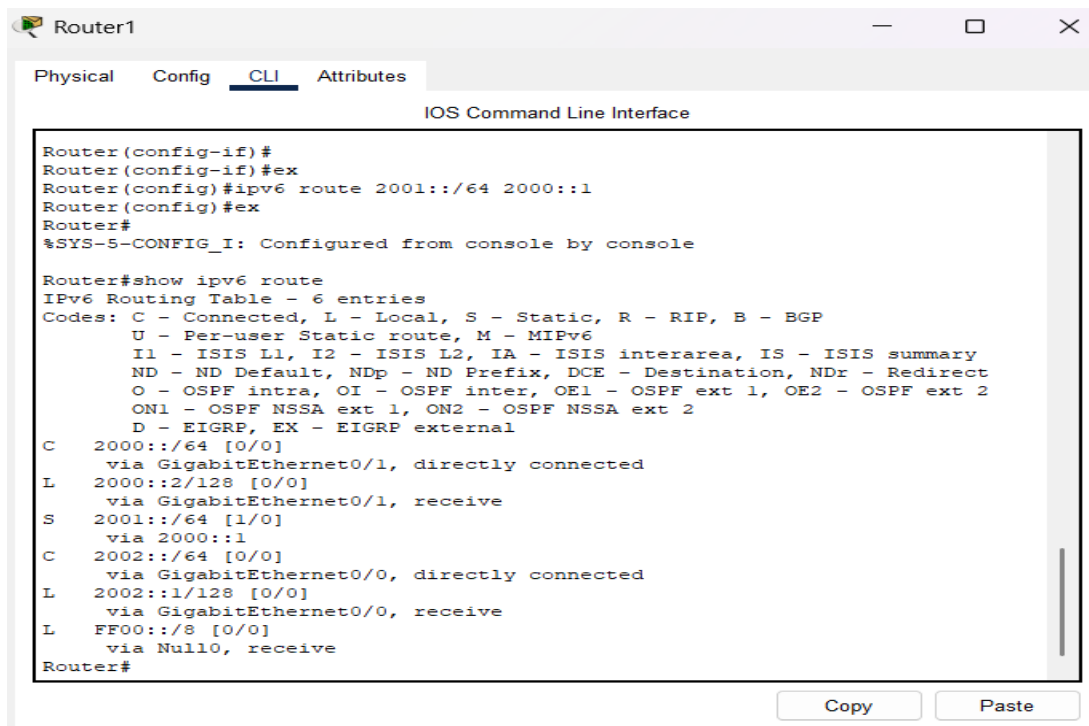


Figure: Viewing static route

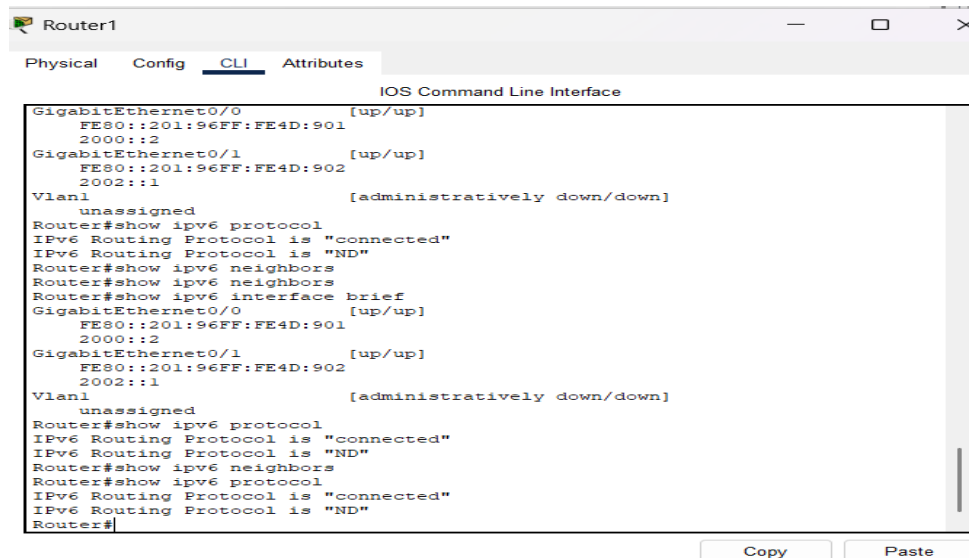


Figure: Viewing IPV6 protocol and neighbors

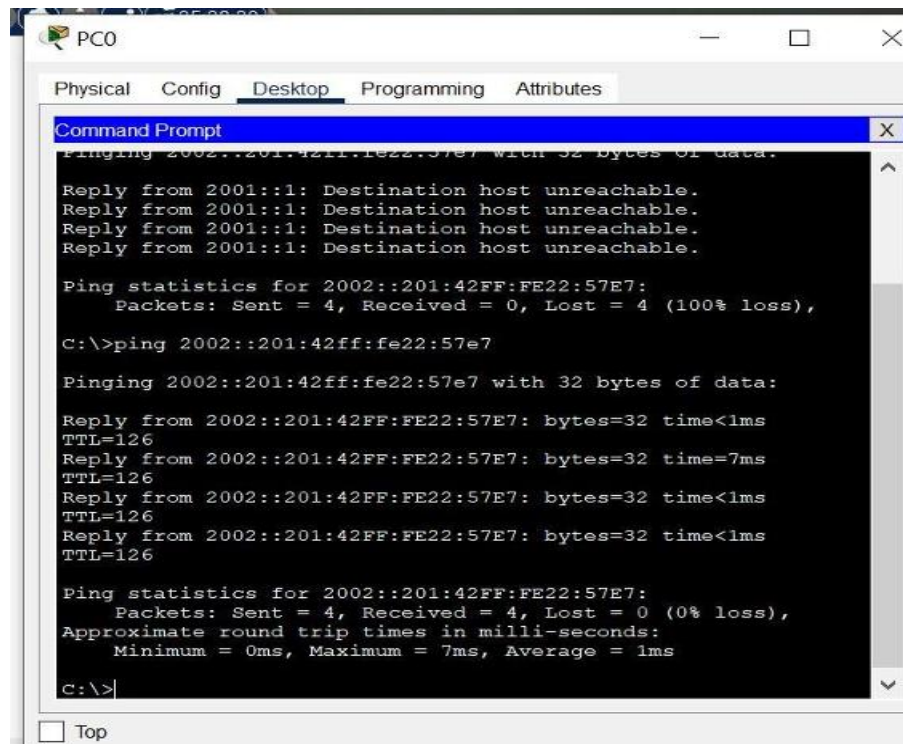


Figure: Testing static route connectivity between two PCs

Lab 4: Dynamic Routing in IPV6: OSPFv3 Implementation

Objective

To understand how system admin can configure and test dynamic routing using OSPFv3.

Theory

OSPFv3 is a link-state dynamic routing protocol designed to support IPv6. It is the successor of OSPFv2, which was used for IPv4. It uses Link-local addresses for neighbor communication (not global IPv6 addresses). It still uses 32-bit IPv4-style Router IDs. It uses Interface-based configuration, No longer uses the network command; OSPF is enabled per interface.

Procedure

In the diagram below, we have used two routers to connect two different networks.

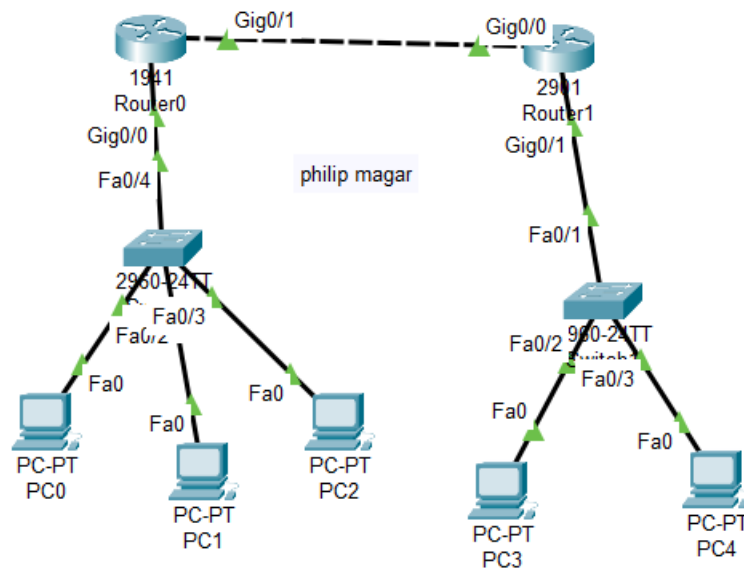
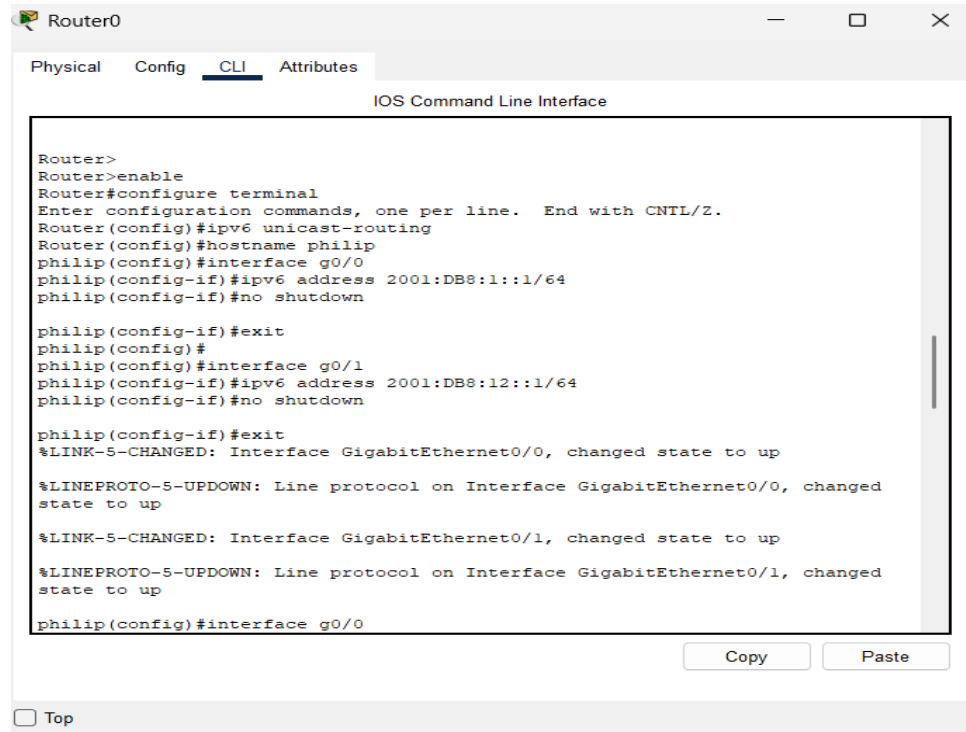


Figure: Network Topology and IPV6 configuration



The screenshot shows a terminal window titled "Router0" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the "IOS Command Line Interface". The terminal output shows the following commands and responses:

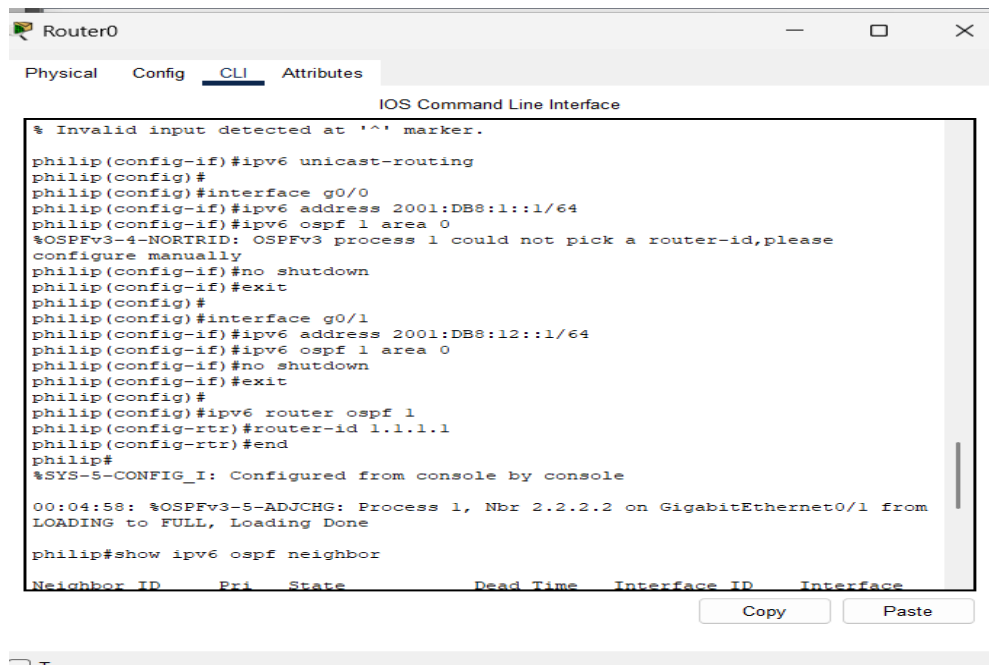
```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ipv6 unicast-routing
Router(config)#hostname philip
philip(config)#interface g0/0
philip(config-if)#ipv6 address 2001:DB8:1::1/64
philip(config-if)#no shutdown

philip(config-if)#exit
philip(config)#
philip(config)#interface g0/1
philip(config-if)#ipv6 address 2001:DB8:12::1/64
philip(config-if)#no shutdown

philip(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
philip(config)#interface g0/0
```

Buttons for "Copy" and "Paste" are visible at the bottom right of the terminal area.

Figure: IPV6 configuration



The screenshot shows a terminal window titled "Router0" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the "IOS Command Line Interface". The terminal output shows the following commands and responses:

```
% Invalid input detected at '^' marker.

philip(config-if)#ipv6 unicast-routing
philip(config)#
philip(config)#interface g0/0
philip(config-if)#ipv6 address 2001:DB8:1::1/64
philip(config-if)#ipv6 ospf 1 area 0
%OSPFv3-4-NORTRID: OSPFv3 process 1 could not pick a router-id, please
configure manually
philip(config-if)#no shutdown
philip(config-if)#exit
philip(config)#
philip(config)#interface g0/1
philip(config-if)#ipv6 address 2001:DB8:12::1/64
philip(config-if)#ipv6 ospf 1 area 0
philip(config-if)#no shutdown
philip(config-if)#exit
philip(config)#
philip(config)#ipv6 router ospf 1
philip(config-rtr)#router-id 1.1.1.1
philip(config-rtr)#end
philip#
%SYS-5-CONFIG_I: Configured from console by console

00:04:58: %OSPFv3-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/1 from
LOADING to FULL, Loading Done

philip#show ipv6 ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
-------------	-----	-------	-----------	--------------	-----------

Buttons for "Copy" and "Paste" are visible at the bottom right of the terminal area.

Figure: OSPF configuration

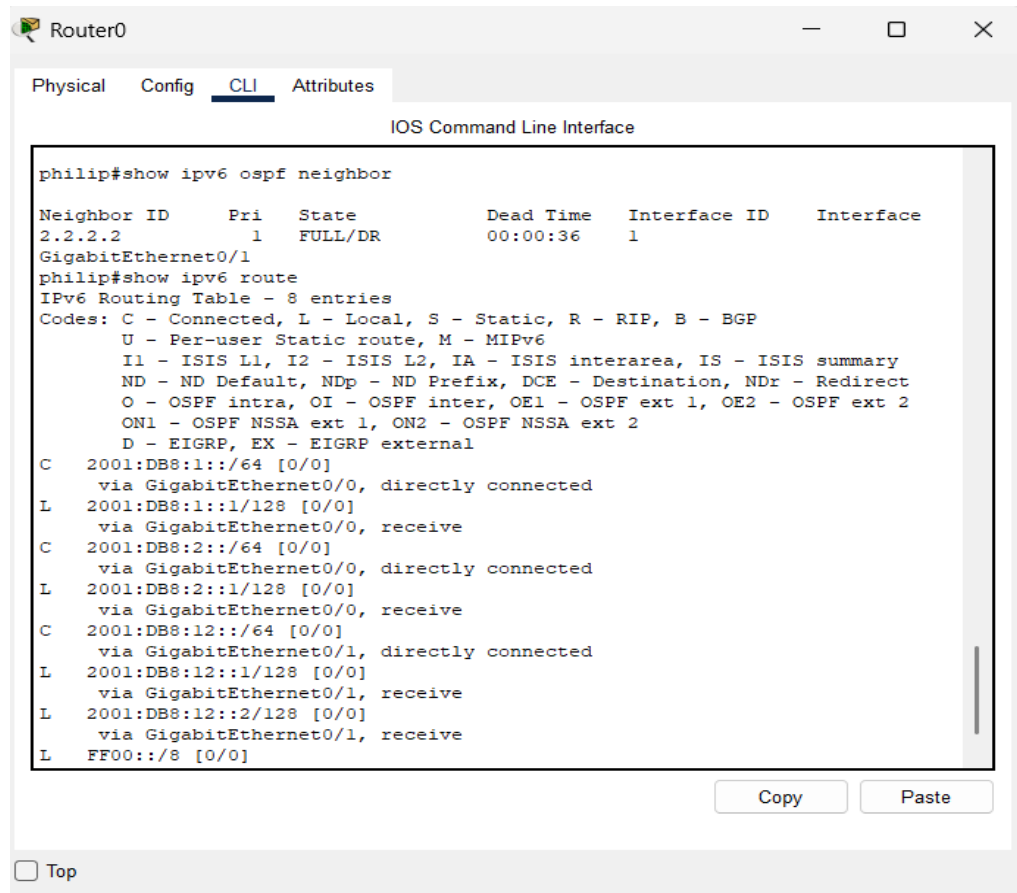


Figure: Visualizing OSPFv3 neighbor and route

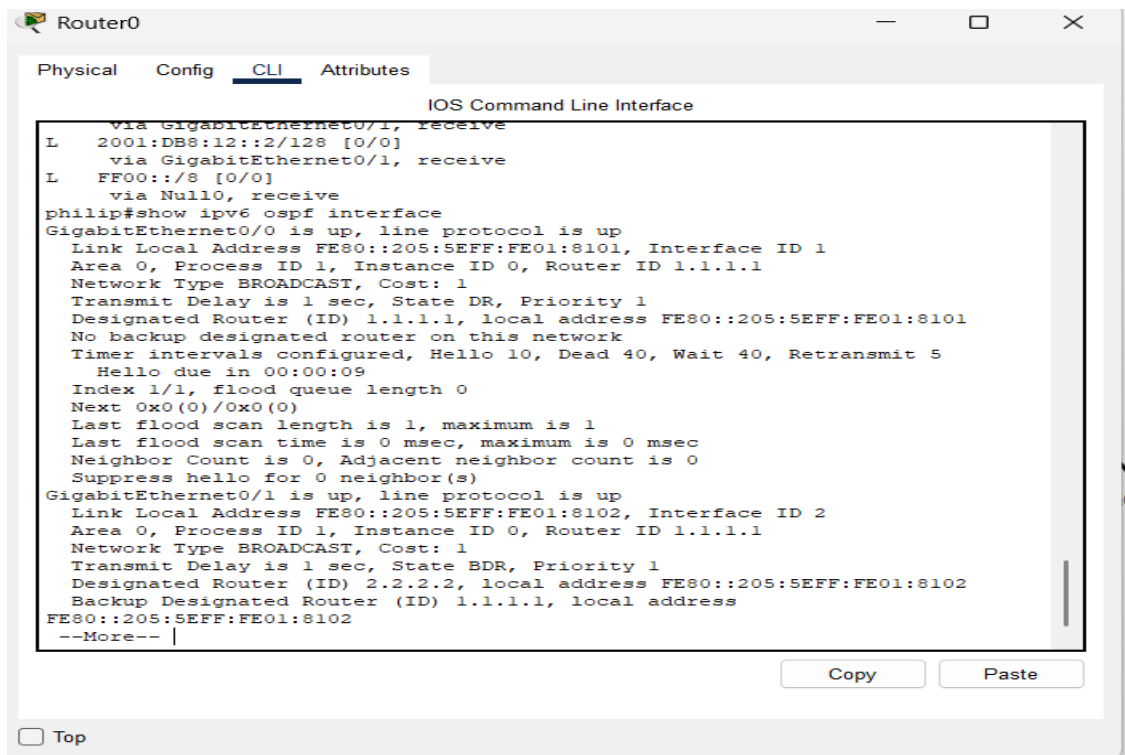


Figure: Visualizing OSPFv3 interface

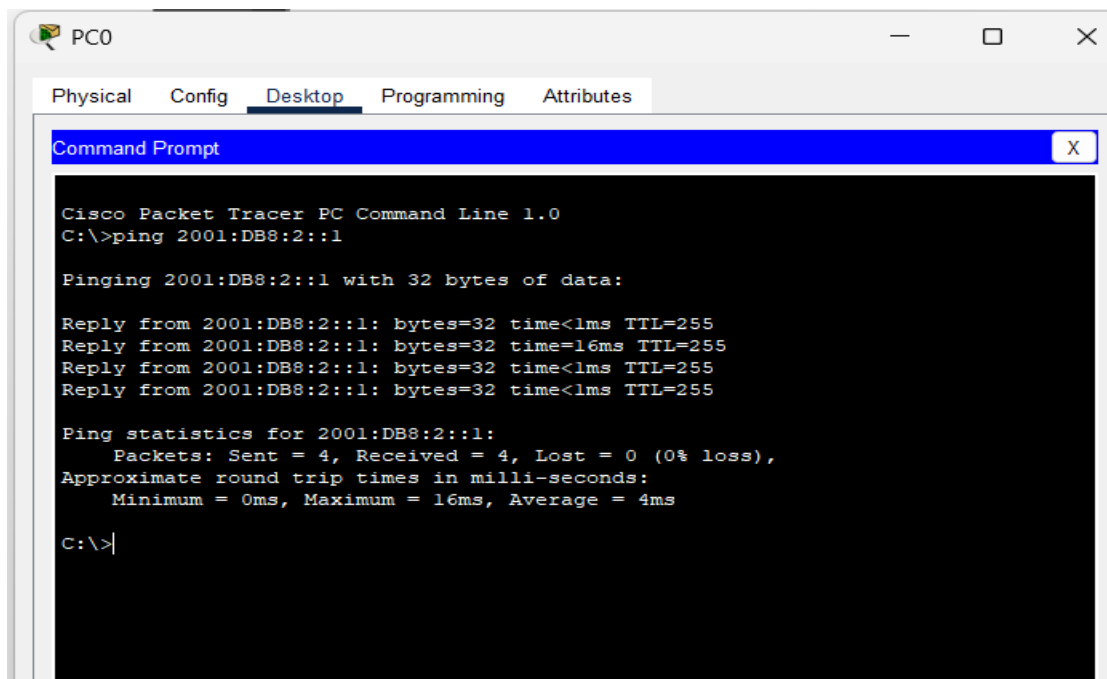


Figure: Testing dynamic route connectivity between two PCs